Approaches to Notating Electroacoustic Music with Live Instruments: Preliminary Results
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ABSTRACT

This paper provides preliminary results from a broad study on the diverse approaches to notating the infinite possibilities of environmental electroacoustic music. This broad study includes a series of case studies on works that integrate environmental soundscapes with electroacoustic music and aims to explore the diversity of methods composers are using today. This paper is primarily reporting on a survey that aims to outline the challenges and possibilities from the perspectives of both the composer and performer and identify methods to engage a wider interest and accessibility in performing mixed music today.

INTRODUCTION

The term soundscape composition defines a genre within electroacoustic music that predominantly uses environmental sounds as source material. As this is one of the fundamental features of electroacoustic music, further ascetical factors are applied to soundscape compositions; such as the relationship to the sonic environment. This particular aspect highlights the fact that sound materials will have a relationship to their source and rarely be perceived in a completely abstract manner. This is a key factor in differentiating soundscape composition, as a large quantity of electroacoustic music prides itself on the Pythagorean "Acousmatic" aesthetic; the apprehension of sound without relation to its source.

Despite this clear difference, soundscape composition still shares many synergies with the field of electroacoustic music, a genre with a particular theoretical strategy developed from the Music Concrete and Electronic Music traditions. The clearest definition of Electroacoustic Music states that the genre is "music in which electronic technology, now primarily computer-based, is used to access, generate, explore and configure sound materials, and in which loudspeakers are the prime medium of transmission" [1]. The use of natural sounds, enriched by technology, allows us to claim that this is the first musical genre ever to place under the composer's control "an acoustic palette as wide as that of the environment itself" [2].

This liberating facet is one of the greatest attractions for the composers as it releases a musical language of infinite possibilities. Despite this liberation, it is the underlying cause into why perception, valuation and particularly analysis of electroacoustic music are extremely difficult tasks. The existing literature has a strong tendency to be extremely complex with demanding terminology obscuring valid theories and the genre has faced considerable negligence predominantly due to contentious issues associated with absent visual referents and infinite sonic possibilities. Western music theory is yet to develop the means to confront such a wealth of sound materials and despite many attempts, a common language or method of analysis does not exist.

THE ACOUSMATIC SCORE

After publishing several papers on the issues of language in electroacoustic music, and an Honours dissertation on Electroacoustic Analysis the author subsequently developed a teaching tool called the 'Acousmatic Score'. This resource employs theories from Albert Bregman, Denis Smalley, Trevor Whishart and Simon Emmerson and enables young composition students to engage with electroacoustic music in an accessible way. The intention of this project was to create a tool to make the genre more accessible and engage a wider audience, and although this addresses some of the obvious problems at a fundamental level, it fails to make a significant impact on truly engaging a wider audience.

This key issue is still driven by exposure and education and in the context of soundscape composition with live instruments it is imperative that it is accessible for performers. This forms the key motivation for a study on the diverse approaches of notating the infinite possibilities of environmental electroacoustic music, with the aim to outline the challenges and possibilities from the perspectives of both the composer and performer and ultimately identify methods to engage a wider interest and accessibility in performing mixed music.

PERFORMING ELECTROACOUSTIC MUSIC

Performing electroacoustic music comes with a range of opportunities and challenges, with most of the challenges revolving around the notion of visual representation; the score and documentation of the compositions. The existing literature frequently documents debates on the strategies concerning the fusion of live instruments and electroacoustic sounds and the way performers relate to electroacoustic music has become a key concern for composers and researchers. In the context of this study we are specifically looking at electroacoustic music featuring soundscapes with live instruments, within this genre a

further division is necessary to discuss fixed and interactive works.

'Fixed' is used to define compositions where the soundscape exists in a fixed form on a CD, DVD or sound file for the performer to accompany. 'Interactive' describes compositions with live electronics, potentially affecting the sound in real time or providing the performer with the ability and control to trigger sound files and different elements of the piece.

The 'fixed' and 'interactive' subgenres come with a range of contentious issues that have contributed towards the genres neglect in contemporary music. Many of these again revolve around issues of accessibility and visual representation. The ultimate outcome is a new model that allows a balance between the extended use of instrumental sources and the infinite possibilities of electroacoustic music in an accessible and sustainable framework.

The role of the score is a critical issue in this context, as early as the 1940 composers were calling for scores to be represented in timbre and texture. The issue of documentation is also a prevailing factor, prior to electroacoustic music the score was essential to guarantee that a composition would survive, and that it could be performed without the composer's presence. Although research has suggested technological innovations provide similar frameworks to the 'score', the fact is a MAX/Msp patch or an Acousmographe diagram is simply not as sustainable, detailed or accurate as traditional music notation. This forms a debatable subject in itself, but unfortunately electroacoustic music is being lost daily purely due to the advancement of technology, ironically the very tool that brought it too life.

LIBERATING THE STAVE

The proposed concept of "Liberating the Stave" is ultimately about developing a unified, accessible and sustainable method for notating electroacoustic music with live instruments. There are multiple research intentions embedded in this concept, the first being to develop an accessible method for the composer, with a particular emphasis on emerging composers engaging with electroacoustic music for the first time. The second intention is developing an accessible method for the performer, this is a key factor, if it is not accessible and engaging it simply won't live past the first performance. The final intention is to develop a sustainable method; specifically not dependent upon technology.

In recent years, new hardware has allowed us pioneering ways to integrate instruments and technology, but due to the complexity it's often a requirement the composer is present to bring the piece to life. When developing models for interactive composition, it's essential the balance between the use of computer-based tools and the relationship between performers and the electroacoustic sounds is thoroughly considered. The technology should be a key to expand the expressiveness of musical language not an obstacle in the performance. So in order to facilitate the intended sustainability the 'art' must always take priority over the technology.

METHODOLOGY

The methodologies for this study are very reliant on qualitative research methods. The first is compiling case studies on significant works in the field and documenting the methods that they use to create visuals representation of their unique sonic languages. The second is a series of comparative analyses between different performances of key works and the third is a survey of performers and composers. Each methodology has produced positive results and while the first will be briefly discussed, this paper is focusing on the preliminary results from the survey.

The case studies have been a key in establishing the most prominent methods and approaches from the perspectives of the composer. Graphic scores are undeniably the most popular notational methods for composers working in this domain; this is not surprising considering western music theory is yet to make any significant contribution to the field. Works such as Ros Bandt's 'Blue Gold' and Leah Barclay's 'Wolf Rock' use a combination of western notation, graphic notation and text information to convey the message to the performer. In this context, the score has been created purely for the purpose of the performer, but in the vein of traditional music notation, now serves as a form of documentation and a tangible tool for analysis. Many of the most prominent mixed works take a similar approach to notation, but the validity of the notation is not necessary a direct result of a successful performance.

Composers may use a graphic notation style open to infinite interpretation, resulting in a work that will be ephemeral and reliant on the performer's improvisation abilities. In soundscape composition, this is often a very liberating and desirable factor, but it means the score is serving more as inspiration as opposed to specific directions.

The comparative analysis methodology is therefore used to test the validity of various notational methods and explore the key issues of interpretation with graphic notation. Though this aspect of the study is revealing some encouraging results, the survey is the most imperative at this stage to begin collating the fundamental data. This survey specifically focuses on the challenges and possibilities of performing electroacoustic music with live instruments.

The survey brings together an array of responses from Australia, New Zealand, Canada, Britain, Scotland, France, Korea, Japan and India with a variety of professional performers and composers, some who regular perform and commission electroacoustic music and some who refuse to perform

electroacoustic music. This study aims to document this area in an honest and candid approach, therefore the identity of the performers is concealed, revealing only their nationality and instrument.

The intention of the survey is to establish why musicians do or don't perform electroacoustic music, identify the perceived challenges and opportunities in performing electroacoustic music, establish a perception on the fixed and interactive debate, identify the most desirable methods for scoring electroacoustic music, and finally to establish an ideal scenario for the score from the perspective of the performer.

RESULTS

The majority of the motivations for performing electroacoustic music are obvious and bear no surprises. "It's invigorating, it's liberating, it's exciting", is a common strand and naturally revolving around the fact that it offers new challenges and opportunities. The explanations for avoiding electroacoustic music have the same predictable results; "it's too challenging, technology is too unreliable, and it's not accessible to tour" being the most prominent.

A Canadian 'cellist who has commissioned a lot of electroacoustic music said "interactive electroacoustic music can continuously vary its response and I can't predict how the computer will react, which is very stressful, when I have a new piece I often have a very organised score for my part and no visual representation of the electroacoustic part so it's this whole other world I have to navigate my way through and it would make it a lot easier to see what is going to happen." This statement from an experienced performer and advocate for electroacoustic music is a clear justification for notating electroacoustic parts. In this context traditional western notation would not be sufficient; the composer could use a combination of graphic notation, sonograms, spectrograms or waveform.

Many performers rate the intimidation of technology as a key deterrent in electroacoustic music, with approximately 87% stating they would be more confident performing electroacoustic music had they had more exposure to this technology in their undergraduate music studies. A New Zealand flautist remarked that she finds "working with the computer very reliable and intuitive, I can do anything with my instrument and the computer won't play a wrong note, it won't go too fast or too slow, it won't lose its place in the score, it won't miss an articulation, and it won't have trouble breathing – it's much better than most performers I've worked with!" She suggested the idea of the composer being more accommodating to the performer is vital, stating "we're here to translate your vision into sound so the more information on the score the better, I find a mix of graphic notation and western notation works well and I like as much information as possible about the electronic part on my score, which is rarely the case." The same performer suggested graphic notation was the most effective method for electroacoustic music, but it's vital that the part is extremely accurate in relation to timing.

The survey questions in relation to the interactive and fixed debate collected a diversity of responses, some quite unexpected for our current music climate. When we consider interactive music, the compositional strategies which include any kind of software for interaction should avoid dependency on the software. It's important to consider 'fixed' music has been active since the conception of electroacoustic music. In the typical performance for an electroacoustic work with instruments from the 1960s onwards performers followed a strict framework with the 'fixed' or prerecorded part. It is quite obvious that this prevents a sense of freedom and a natural interaction between the performer and the electronic part, hence the progression to interactive electroacoustic music.

In the context of fixed music, even highly innovative and complex parts will never be affected or modified by the performers' decisions. The technological developments in the progression to interactive music now create infinite possibilities of what we can do in real time, this also facilitates a sonic dialogue and interactivity between the performer and electronic part. Interactive electroacoustic music is much more flexible and obviously allows the performer a greater degree of freedom, but as the diverse survey results reveal, this does not necessarily mean it is the preferred method.

Although 'interactive' electroacoustic music is more flexible, it is important to take into consideration the requirements of the composition, and certainly not employ interactivity for the sake of it. The intention must always come from within the piece. Performers also suggest when using software such as MAX/msp, an understanding of the patch and exactly how it works is a highly desirable factor.

Writing a MAX/msp patch is a complex notational language in itself, and many composers take an extremely different approach. The notion of creating a graphic score and a full written documentation of the patch creates accessibility for the performers and it also creates the sustainability for the composition, if a full written documentation of the patch exists it can ultimately be recreated in other platforms when the next software advancements materialize.

The responses from the surveyed performers all offer valid and diverse opinions on this subject. An Australian percussionist said she loves interactive music, responding with; "knowing I have the Freedom to work with the part I feel like I'm so much more involved in the piece and engaged with it – playing with a fixed track is like playing with a backing track I feel like I'm just playing along and feel quite restricted and trapped in the part – even just a couple of triggers are enough to give the performer that sense of control."

This comment is a clear indication that the control factor in interactive music is highly appealing.

A prominent British electroacoustic composer provided some strong responses to interactive music in the survey, stating that he believes "MAX/msp and interactivity is not at all sustainable, the works disappear, we have no documentation and it's a frightening position for the composer when we leave our selves so vulnerable to technology, I am not interested in MAX and can explore all I need with my fixed resources in the studio with much better quality." An acclaimed Australian performer and composer said she will not play interactive music commenting "you have no control, the quality is often terrible and things always go wrong". She would much rather play with a professionally mastered CD, or 'fixed' part and know that the piece will work without the unpredictability and stress of interactivity.

A Chinese violinist said he only plays interactive works as the spontaneity and excitement is what drew him to the genre. An Australian Clarinetist said he would rather play fixed works as they are so much easier to tour and come with less technical problems. The responses to the interactive and fixed debate continue with inclusive results, overall 38% of the surveyed performers prefer working with fixed composition, 33% preferred interactive compositions and 29% were happy to work in both areas. The overall responses show that the key is to give the performers a sense of control, create a balance between the live instruments and electronics and facilitate that process in a framework that is accessible to tour without too much technical assistance. In this context, it essentially becomes irrelevant if the composition is fixed or interactive, it is completely dependent on what the piece requires. Many composers are now exploring the possibility of creating multiple versions of the same work to facilitate this accessibility. 'Wolf Rock' for example by Leah Barclay exists as an accessible fixed version that is easy to tour and an interactive version for when the time and resources are available for a full performance.

The concept of establishing an ideal score from the perspective of the performer shows that the majority of artists participating in the survey simply require more information and a visual representation of the electronic part. Out of the performers surveyed 93% find graphic notation the most effective method, and 83% say they would be interested in exploring new interactive methods that use tools such as animation.

It is evident that to 'liberate the stave' a combination of graphic notation, western notation, sonograms, spectrograms, and tools such as GRM's 'Acousmographe' is required to create 'perceptual' scores. There are many other examples of digital tools available on the market to create graphic representation of sound. Yet, many performers surveyed suggest a composer's own graphic notation tells them so much about the piece.

An Australian guitarist says "Graphic notation speaks to me in a way western notation doesn't, I feel more connected to the gesture and energy the composer is creating and I feel a sense of freedom working with the electronics in this context". A number of performers share a similar perspective, and emphasize the fact that notation is essentially a fixed physical form of symbols that represent a musical action. The great revelation of the notational experiments of John Cage, Earle Brown, and Morton Feldman challenges the very meaning of these symbols. Their intentions were to point to the infinite possibilities of any interpretation of a notation and to engage the performer in the creative process of the piece. This form of graphic notation can still be created in digital software but the results of the survey suggest it does appear to be more effective in the composers own hand.

CONCLUSION

This paper has provided some preliminary results in the methods of notating live instruments with electroacoustic music, and more specifically environmental soundscapes. It is evident a full documentation of the work, in both visual and text formats is essential for sustainability and accessibility and a fusion of existing notational tools and innovative methods is essential in creating this framework.

The final outcome of this research is combining the results from the case studies, comparative analyses and survey and developing a unique graphic notation for environmental electroacoustic music. This model will place a strong focus on accessibility and sustainability, and facilitate interdisciplinary collaborations fueling the possibilities for environmental electroacoustic music to have implications across a diversity of fields.

REFERENCES

[1] Emmerson, S., & Smally, D. (2001). Electro-acoustic music. In The new Grove dictionary of music and musicians (Vol. 8, pp. 59-67). London: Macmillan.

[2] Emmerson, S. (1986). The relation of language to materials. In S. Emmerson (Ed.), The language of electroacoustic music (pp. 17-39). Basingstoke, UK: The Macmillan Press.